

Device Risk Value Prop Analysis

System Accuracy Measures and Monetary ROI

Anonymous Credit Provider

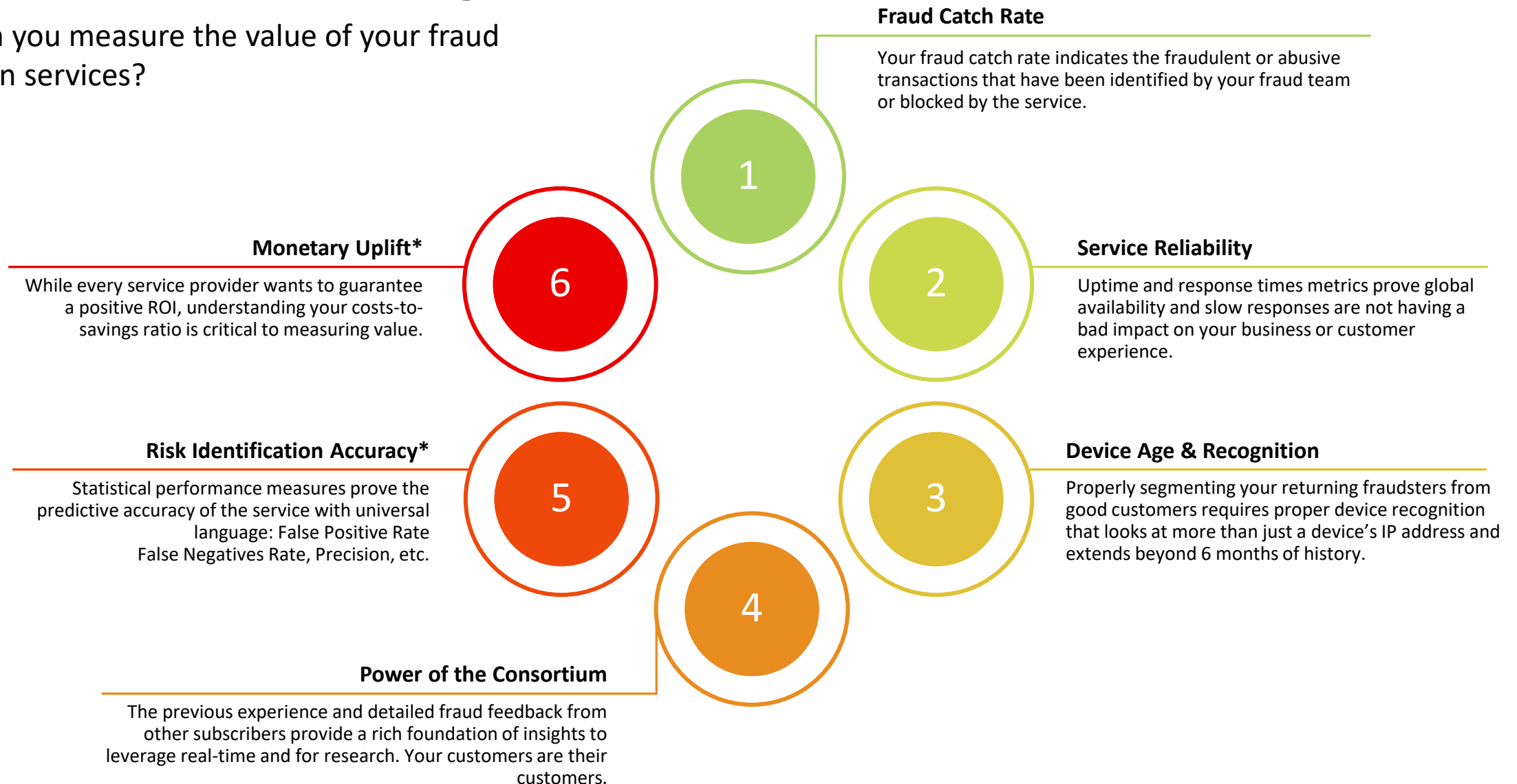
Industry: **Financial Services** | **Credit Card**

Analysis Window: **Q4 2020 & Q1 2021**



Options for Determining Value

How can you measure the value of your fraud detection services?

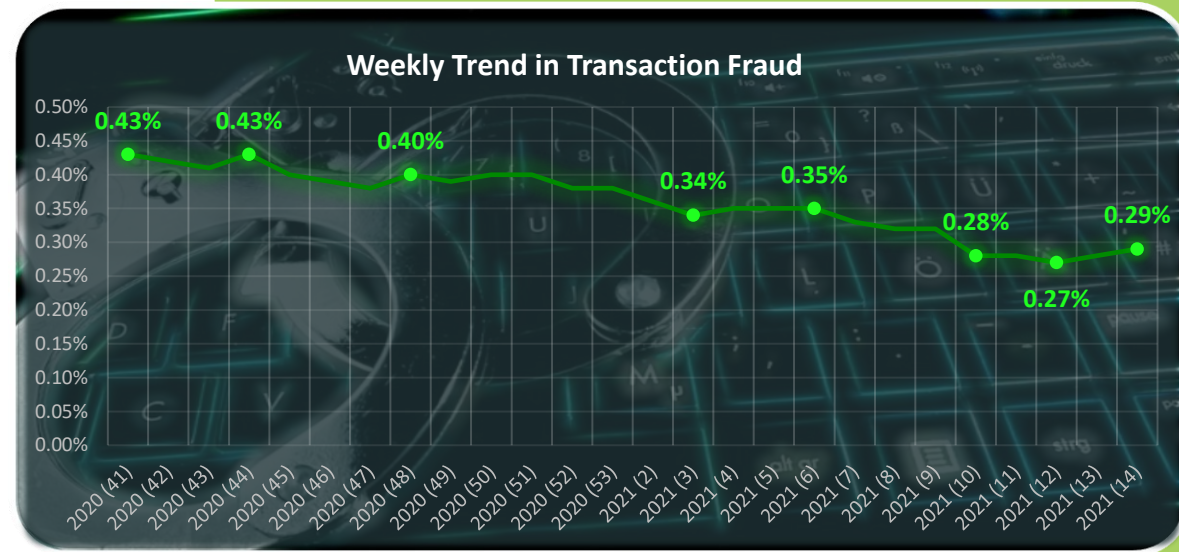


0.36%

Fraud Catch Rate

Your fraud catch rate reflects the fraudulent or abusive transactions that have been identified by your fraud team or blocked by the fraud detection service.

Financial Industry Average: 1.40%



Analysis Window: Q4 2020 & Q1 2021

Cyber Attacks and Automated Activity

920,810

Cyber-Related
Attacks

5.0%

Confirmed Fraud
or Abuse Rate

● **193,523**

Transactions with High Botnet IP Risk

Risk Score Over 7 (out of 10)

● **174**

Transactions Triggering a Proxy or VM in Use Rule

Pierced Proxy shows Mismatch (Excluding Anonymous Proxy)

● **721,203**

Transactions with Geolocation Obfuscated

Anonymous Proxy

● **0**

Transactions Triggering TOR Network Rule

IP is an Active Node in a TOR network Today

● **5,910**

Transactions Utilizing a TOR Network IP

IP Registered in a TOR Network Anytime

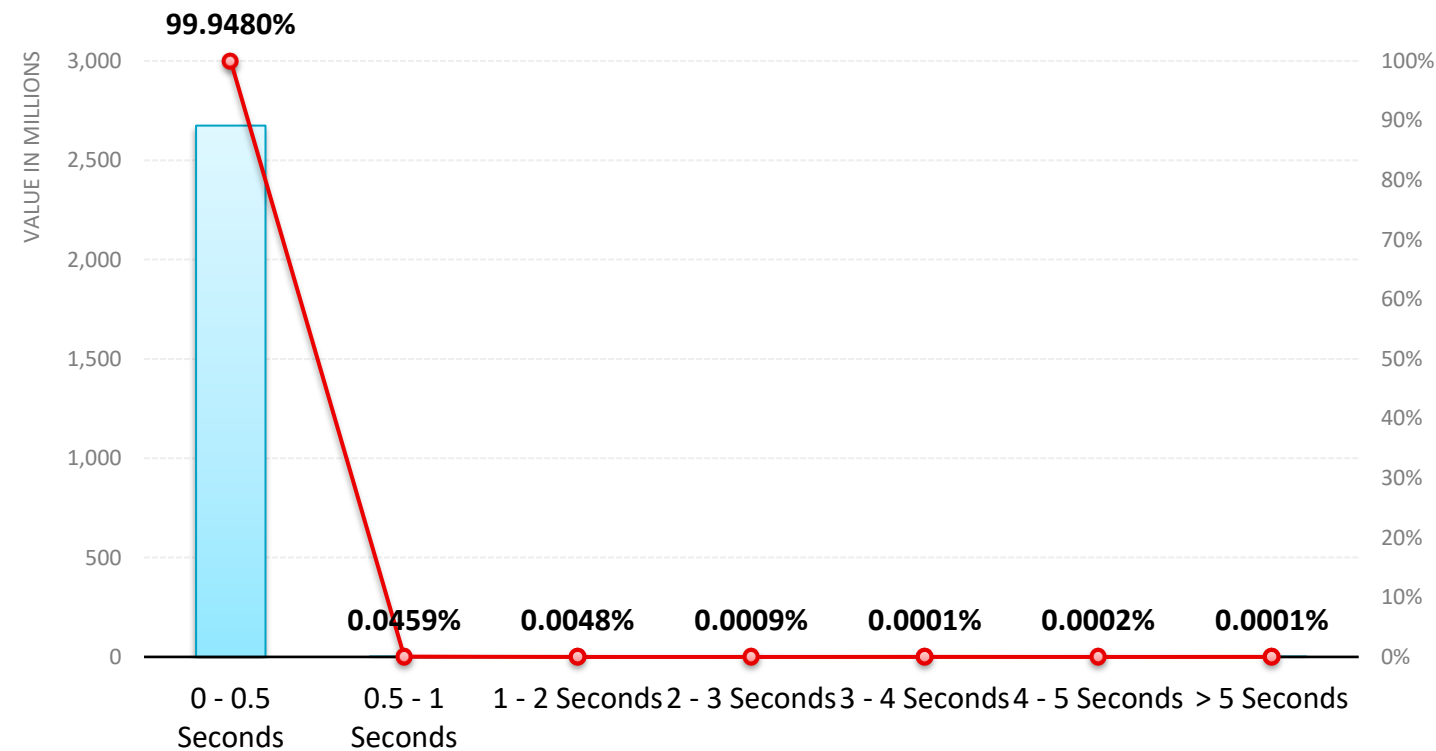
2

Service Reliability

Uptime and response times metrics show that global availability and slow responses are not having a bad impact on your business or customer experience.

Response Time Buckets

Global Activity Over Past 90 Days



99.948%

of transactions receive a service response within 0 - 0.5 seconds

Bucket	Transactions	Rate
0 - 0.5 Seconds	2,674,359,470	99.9480%
0.5 - 1 Seconds	1,227,077	0.0459%
1 - 2 Seconds	129,540	0.0048%
2 - 3 Seconds	23,336	0.0009%
3 - 4 Seconds	3,758	0.0001%
4 - 5 Seconds	4,542	0.0002%
> 5 Seconds	3,195	0.0001%

New Device Rate
3.11%
7,685,271

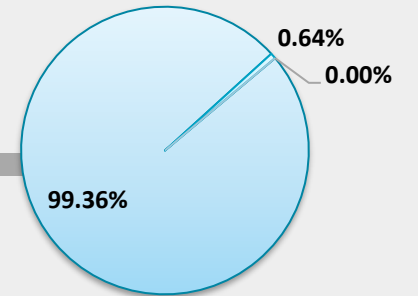
Returning Device Rate
96.89%
239,677,907

Device Capture

First and third party collection methods are available to ensure the user's privacy settings do not thwart device collection methods.

Breakdown of No-Device Reasons

■ No Blackbox ■ Invalid BB ■ Suspect



Device Recognition

New Device Rate reflects the number of devices that were new to Transunion when first interacting with your brand. Returning devices were re-recognized by the service and their histories are available for new risk decisions for over 2 years.

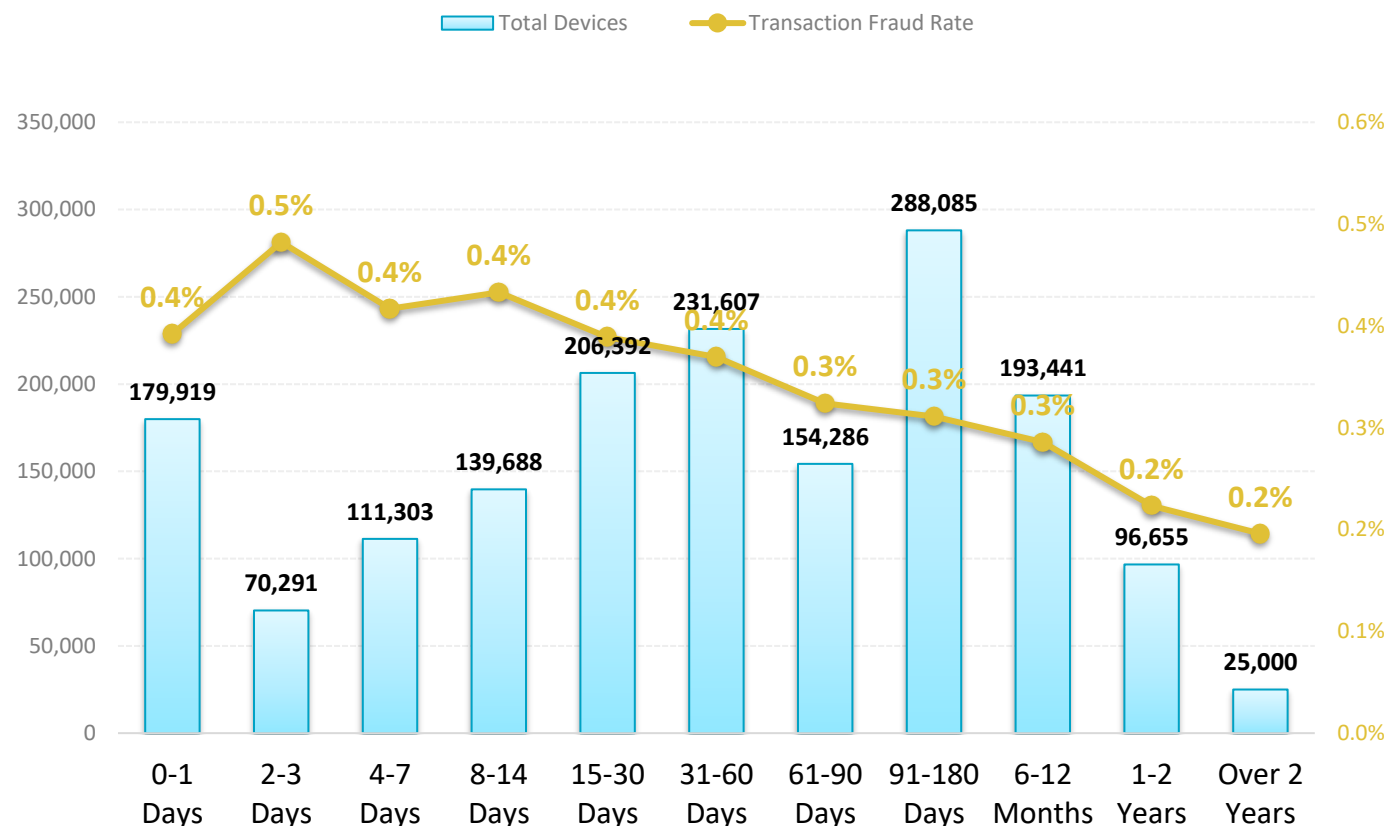
Device Capture Rate
99.72%
247,363,178

No Device Rate
0.28%
683,280

Value of Device Retention

For devices that were new to the subscriber at the time of the transaction, this chart summarizes the average number of days Transunion had exposure to their device before it visited their site. While the device was previously unseen by you, our lengthy device retention allows us to provide history and reputation for over 2 years.

New-Device Age Distribution
Age of Devices Previously Seen by TU at Transaction Time





Power of the Consortium

Consortium Fraud-Catch Opportunities exist when other subscriber place evidence on crossover devices interacting with your business where you have NOT placed that type of evidence on the device.

Crossover devices with a history of nefarious behavior provide rich insights about devices that you have not yet identified as fraudulent/abusive.

Evidence Placed	Direct Devices Identified (Client)	Total Devices with <u>Only</u> Consortium Evidence	Total Devices with <u>Only</u> Industry Peer Evidence	Total Transactions	Transactions Performed by your Devices with Real-time Result		
					ALLOW Results	REVIEW Results	DENY Results
ACH/Debit Fraud	135	1,519	1,489	25,695	242	3,319	22,134
Application Fraud - 3rd Party	19,717	2,706	987	23,647	1,199	3,571	18,877
Account Take-Over/Hijacking	17,688	2,762	1,929	48,339	767	22,456	25,116
Application Fraud - 1st Party	27	8,588	8,560	137,038	779	89,546	46,713
Claims Fraud - 3rd Party	18	35	15	420	4	176	240
Claims Fraud - 1st Party	14	5	5	7	-	2	5
Credit Card Fraud	10	16,041	542	235,002	4,311	22,651	208,040
Potential Fraud	1	17,743	13,641	254,628	2,028	42,342	210,258
Identity Mining/Phishing	2	71	52	994	79	168	747
Insufficient Funds	1	515	118	5,709	8	628	5,073
Fraud - Other	1	15,176	11,877	191,645	1,591	17,620	172,434
True Identity Theft	1	7,792	5,322	92,717	1,672	9,554	81,491
Synthetic Identity Theft	1	1,660	1,648	30,408	404	8,677	21,327
Grand Total*	37,616	122,328	62,857	1,823,887	15,639	284,166	1,524,082



Who Found Fraud Knew First?

Power of the Consortium

Crossover devices with confirmed fraud identified by other members of the consortium. The list shown is limited to relevant industries with similar fraud concerns (evidence types) as you.

Evidence Placed	You and another Subscriber Placed Evidence on the <u>Same</u> Device		Credit One Ignored Evidence	Credit One Never Placed Evidence	Grand Total
	Other Subscriber Found Fraud First	Credit One Found Fraud first			
Financial Services	358	230	34	22,489	23,111
Short Term Lending	25	13		5,281	5,319
PrePaid Cards	72	36		4,936	5,044
Money Service	134	62		4,076	4,272
Retail Banking	93	89	34	2,868	3,084
Banking	6	6		1,687	1,699
Online Credit	4	11		1,407	1,422
Credit Card	6	11		1,194	1,211
PSP	13			514	527
Operator	2	2		201	205
Federal Government	1			147	148
Credit	2			141	143
Credit Union				34	34
Merchant				3	3
Retail	67	35	1	65,429	65,532
Merchant	49	31	1	61,537	61,618
Online Marketplace	7	4		1,695	1,706
Dating	9			1,185	1,194
B2B	1			781	782
Forex	1			209	210
Credit Card				21	21
Operator				1	1
Risk Management		1		213	214
Service Provider		1		125	126
OEM				81	81
Money Service				7	7
Grand Total *	425	266	35	88,131	88,857

Analysis Window: Q4 2020 & Q1 2021

Risk Identification Accuracy

Statistical Performance Measures that Prove the Predictive Accuracy of the Service

5

Risk Identification Calculations

Statistical performance measures show the predictive accuracy of the service in identifying risk using common terminology. Based on your evidence feedback, we can measure the frequency in which the fraud outcome matches the risk identified.



99.6%

Accuracy (ACC)

$$ACC = (TP + TN) / (TP + TN + FP + FN)$$

Accuracy is the proportion of correct risk predictions made by the classification system or machine learning model.

Low accuracy rates indicate that the system is not correctly identifying enough of your returning fraudsters and requires more inputs to learn the proper risk response and increase predictability.



0.2%

False Positive Rate (FPR)

$$FPR = FP / (FP + TN)$$

The false positive rate indicates the probability of raising a false alarm for a good customer and creating unwanted friction at the time of the transaction.

Higher FPR values typically indicate infrequent evidence placement or inconsistent manual review practices to correctly identify fraud after a risk flag and follow through with feedback.



87.4%

False Negative Rate (FNR)

$$FNR = FN / (FN + TP)$$

The false negative rate reflects missed fraud catch opportunities where a confirmed fraudster was not flagged as risky based on current business rule configuration.

Higher FNR values typically indicate that the business rules are not tuned frequently enough to keep in step with changing fraud tactics and emerging trends. Unwanted behaviors are not being targeted.

5

Risk Identification Measures

Statistical performance measures show the predictive accuracy of the service in identifying risk using common terminology. Based on your evidence feedback, we can measure the frequency in which the fraud outcome matches the risk identified.

True Positives (TP)

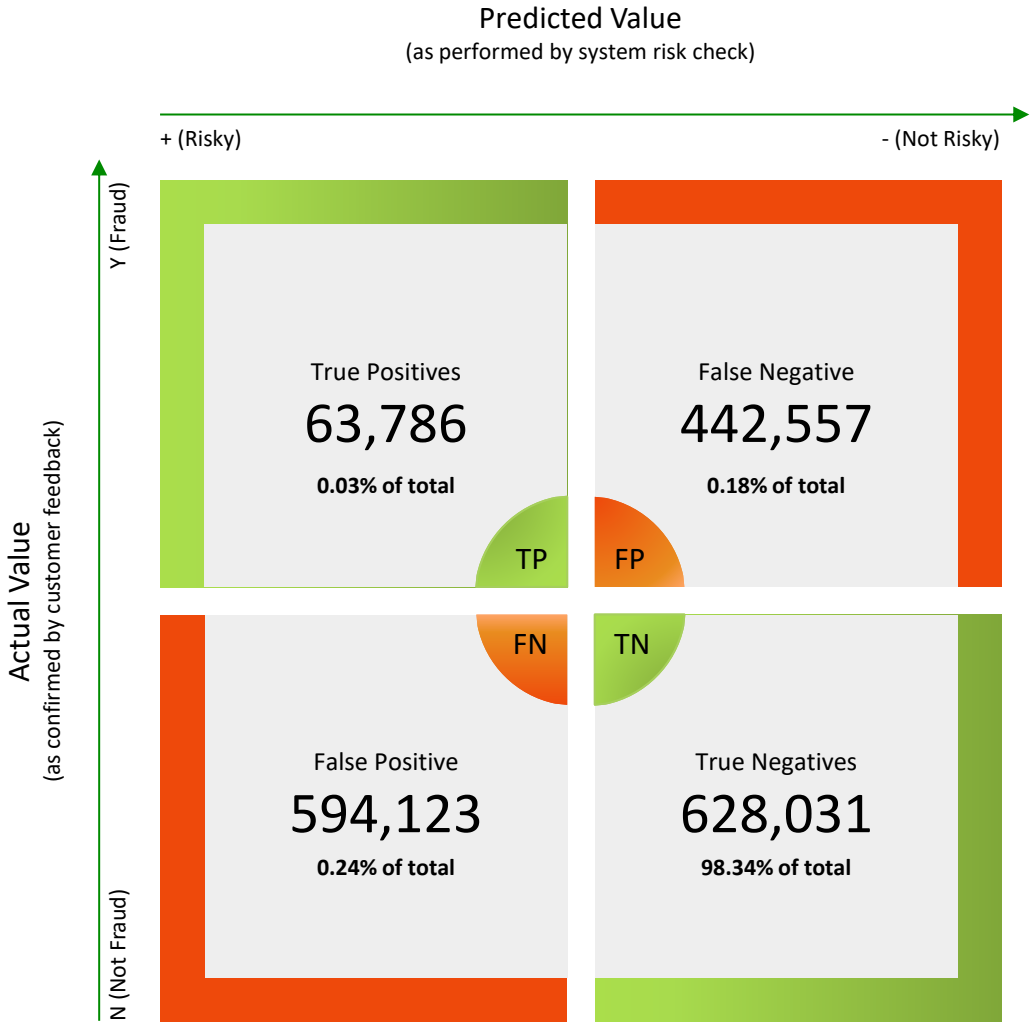
The service got it right! A transaction, customer, or device was flagged as risky, fraudulent, or abusive and was confirmed as such through client feedback or evidence placement.

The service returned a REVIEW or DENY at the time of the transaction and today there is still evidence on the account or device. Returning fraudsters that are recognized as the same device are also included in this set of positive predictions.

False Positives (FP)

The service got it wrong. Customer Friction! A transaction, customer, or device was flagged as risky, fraudulent, or abusive but turned out to be good customer based on evidence today.

At the time of the transaction, the service returned a REVIEW or DENY and there was no existing evidence on the account/device (i.e. not a returning fraudster), but today there is still no evidence to confirm the suspicion was correct.



False Negatives (FN)

The service got it wrong. Missed Fraud! A transaction that was NOT flagged as risky but the account/device was later reported as fraudulent or abusive by the client or through evidence placement.

The service returned an ALLOW at the time of the transaction and this was not a returning fraudsters, but there is now evidence on the account or device today. Therefore, the service missed an opportunity to flag that activity as fraudulent.

True Negatives (TN)

The service got it right! A transaction, customer, or device was NOT flagged as risky and was confirmed as a good customer by the client or no evidence was placed to date.

The service returned an ALLOW at the time of the transaction and neither the account nor device was associated with evidence, and still today there is no negative feedback (evidence) on that account or device.



Types of Missed Fraud

False Negative transactions where the service did not alert you to fraud or risk but you later placed evidence on the account or device.

Evidence Placed	Devices with Direct Evidence	Accounts with Direct Evidence	Devices with Indirect Evidence	Accounts with Indirect Evidence
ACH/Debit Fraud	72	8,577	1	7,475
Application Fraud - 3rd Party	90	2,139	1	1,593
Account Take-Over/Hijacking	1,013	13	0	0

Analysis Window: Q4 2020 & Q1 2021

Monetary Uplift and ROI

Estimated Return on Investment Based on Usage and Costs

Assumptions in Preparing ROI

Determining your exact return on investment is impossible without the exact fraud losses and monetary gains associated with each account. However, we can estimate your costs-to-savings ratio based on how you are using the service, your investment costs, and an average fraud loss amount per account, which will applied to all fraudulent accounts identified by the service.

This ROI is an estimate of uplift using inputs provided by you.



Service Overview:

- All costs used in calculations were based on manual review and fraud costs you provided
- Transaction volume based on traffic from all integration points



Billing:

- Actual invoiced amounts during the analysis timeframe were used for calculating billing totals
- Where applicable, a blended rate was used for partial-month transaction costs not yet invoiced



Workflow:

Initial Focus

- **100%** of REVIEW responses and **5%** of DENY responses are manually reviewed
- **95%** DENY responses lead to a real-time stop
- Additional fraud prevention service calls are made following TruValidate for a total of **\$1** per transaction

Cost Inputs for ROI Model

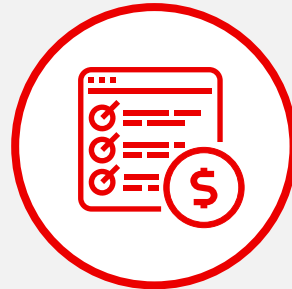
**Average Cost of
Fraud per Account**



200.00

Value provided by you in ROI
questionnaire for average fraud
loss per account

**Average Cost of Manual
Review per Transaction**



4.85

Value provided by you in ROI
questionnaire for operational cost
for each reviewed transaction

**Additional Service Fees
incurred after
TruValidate**

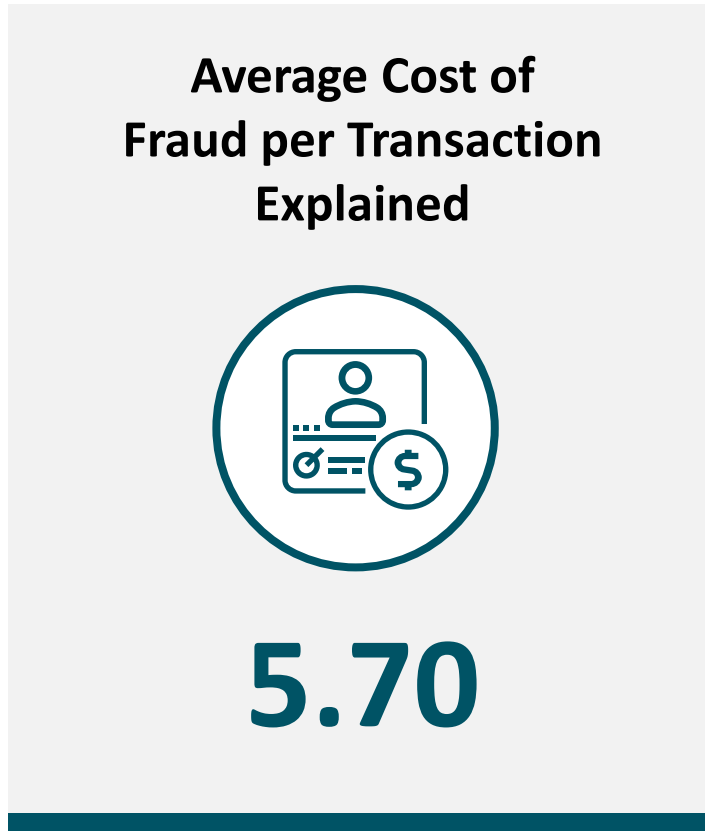


1.00

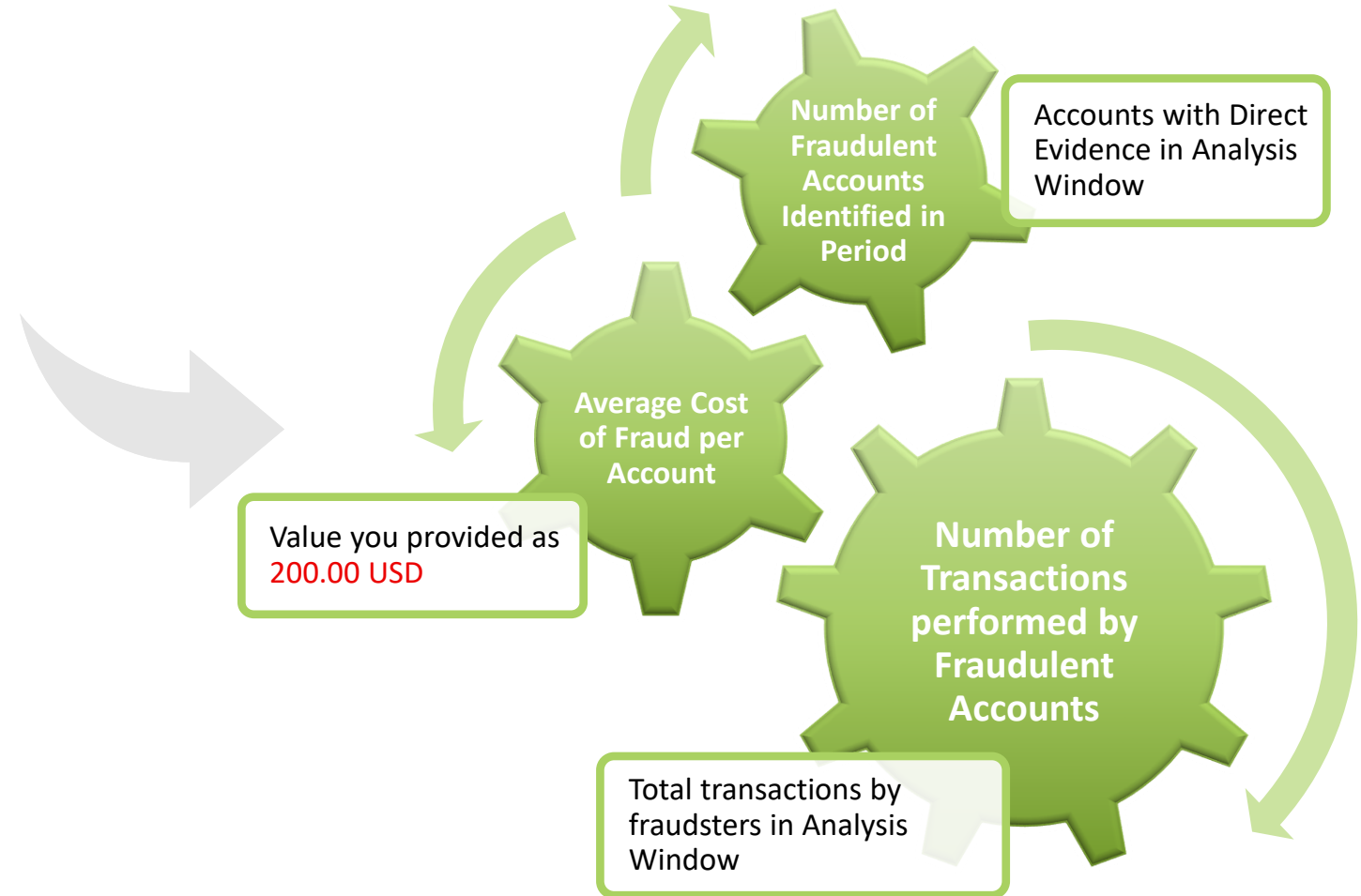
Value provided by you for
technology costs from other
fraud prevention providers

Cost Inputs for ROI Model

Special Calculation based on Cost Inputs

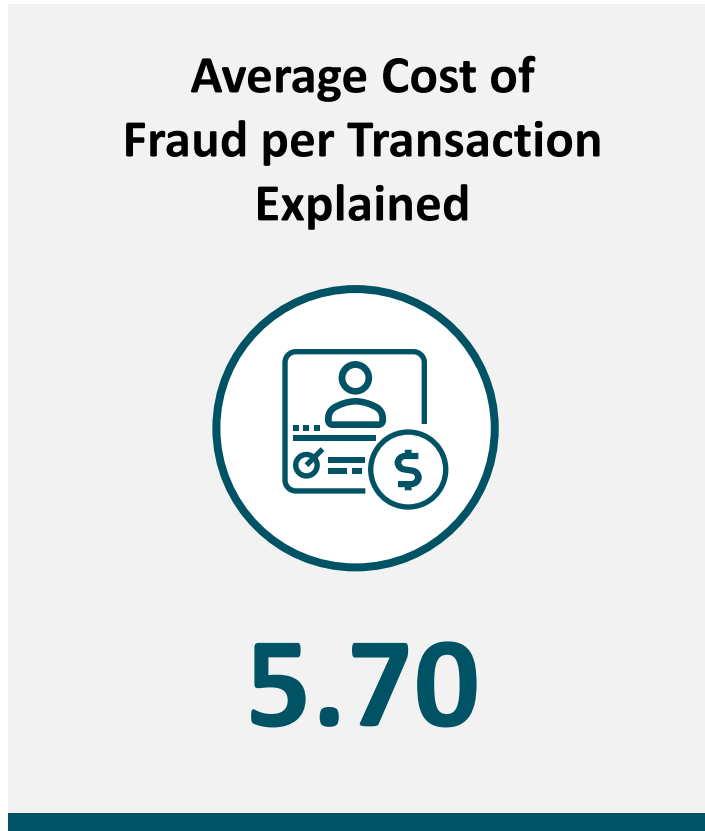


Calculated from Average Fraud Loss per Account and the actual fraud identified in the analysis window

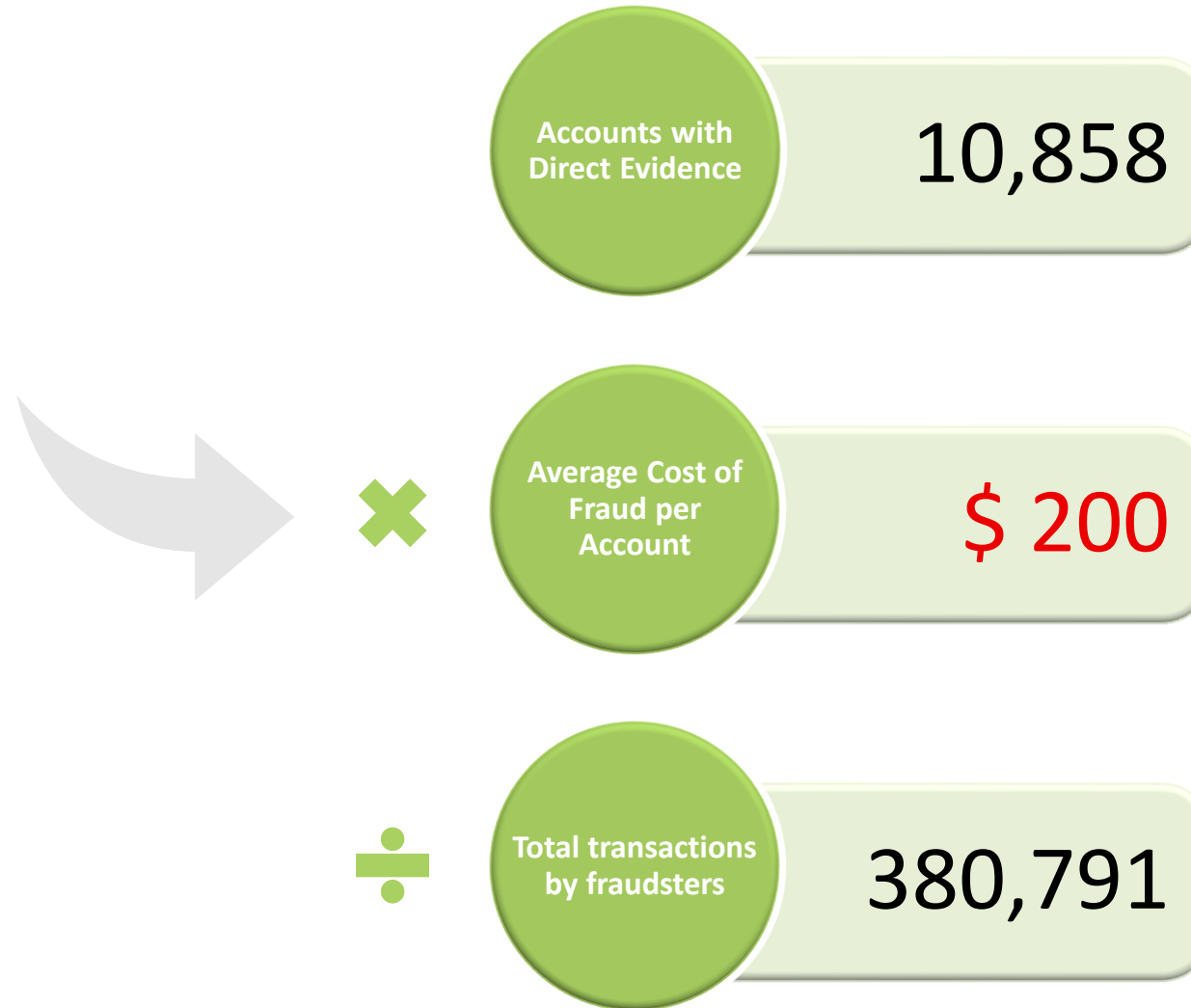


Cost Inputs for ROI Model

Special Calculation based on Cost Inputs

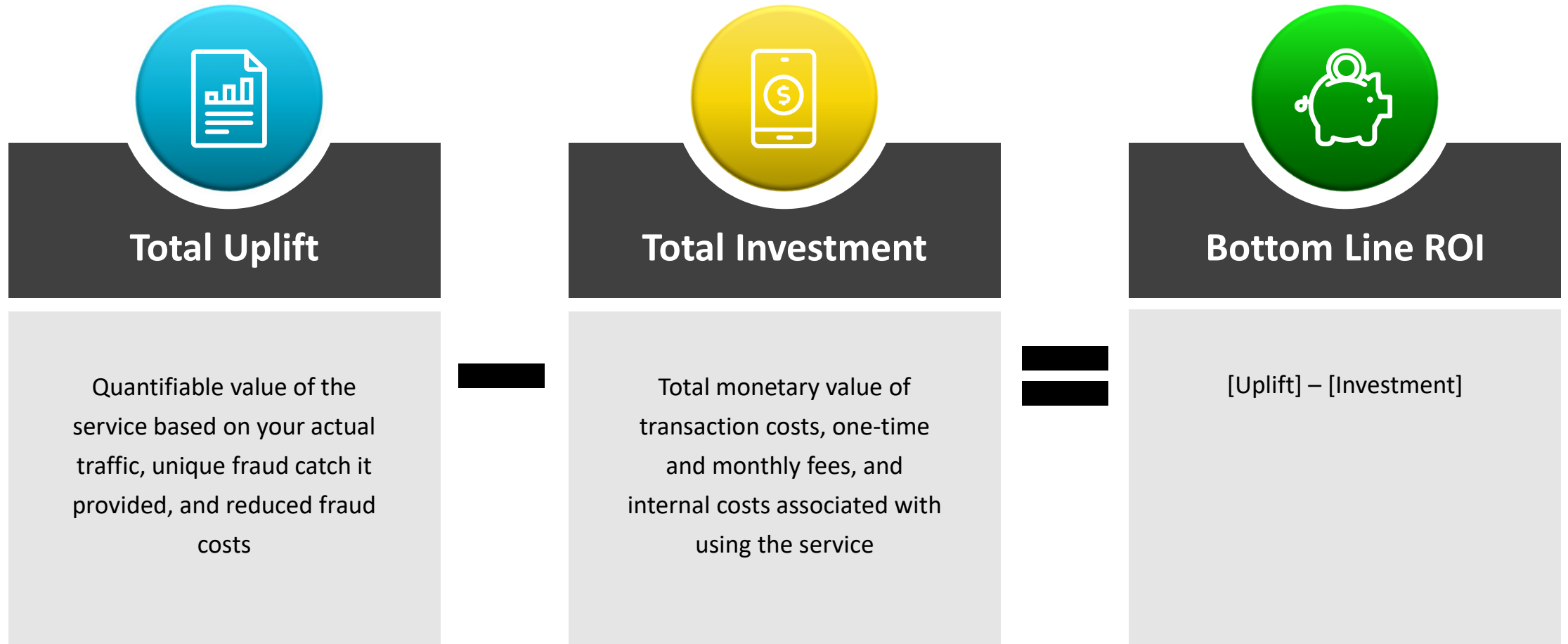


Calculated from Average Fraud Loss per Account and the actual fraud identified in the analysis window



The Formula for Calculating ROI

The Big Picture





Components of Investment

Costs Incurred while using TruValidate Device Risk Service



Billed Transactions

- ✓ Invoiced Amount for Transaction Volumes over Full Months
- ✓ Calculated billing for partial month(s) using best Blended Rate



Monthly Fees

- ✓ Premium, On-Call Support from award-winning Client Services
- ✓ Custom Reports Delivered Regularly to Designated Recipients



One-Time Fees

- ✓ One-time Integration Fees Billed at the Start of Service or at the beginning of Proof of Concept trial period
- ✓ One-time Custom Report Fee



Personnel Costs

- ✓ Transactions that led to manual review based on your current workflow
- ✓ Considers both REVIEW and DENY responses from TruValidate



Components of Uplift

Uplift Unique to TU Services that Complement your Existing Fraud Team Efforts



Uplift by Indirect Accounts Associated with Evidence

- ✓ You confirmed fraud by placing direct evidence on an Account or Device
- ✓ iovation device linkage identified additional Accounts using same Device
- ✓ iovation propagated evidence to associated accounts
- ✓ You saved the cost of fraud from linked accounts with indirect evidence

Uplift by Identifying Confirmed Fraudulent Accounts

- ✓ iovation rules triggered account to be reviewed (suspicious behavior pattern)
- ✓ You reviewed the account/device history with your brand and confirmed fraud
- ✓ iovation prevented current transaction with risky customer and device
- ✓ You did not incur the cost of fraud for each of these accounts

Components of Uplift

Uplift Unique to TU Services that Complement your Existing Fraud Team Efforts

Uplift by Preventing Real-time Fraudulent Transactions

- ✓ iovation deemed transaction suspicious and fits known fraud behavior pattern
- ✓ iovation sent a DENY response
- ✓ Your systems prevented fraudster from completing transaction
- ✓ You did not incur the cost of fraud for each of those transactions

Uplift by Reduced Review Costs

- ✓ iovation deemed transaction suspicious and fits known fraud behavior pattern
- ✓ iovation returned a DENY response
- ✓ You saved the cost of manual review of those accounts



Components of Uplift

Uplift Unique to TU Services that Complement your Existing Fraud Team Efforts

Uplift by Reduced Technology Costs

- ✓ Innovation sent a DENY response
- ✓ Your systems exited transaction workflow and prevented additional service calls
- ✓ Your systems prevented fraudster from completing transaction
- ✓ You saved the cost of calling additional fraud services to verify identity, check balances, process payments, etc.

ROI: Current Utilization

TruValidate Device Risk service for Q4 2020 and Q1 2021

TOTAL COSTS (INVESTMENT)	Quantity	Unit Price	Amount
Billed Amount for Period	238,396,196	0.0038	932,455
One-time Fees / Monthly Fees	6	0.00	0.00
Manual Review Costs	826,784	4.85	4,009,901
Investment Over 6 Months			\$ 4,942,356

Billing Month	Transaction Volume	Per Query Price	Billed Amount	Billing Note	Monthly Fee
OCTOBER 2020	24,599,629	0.65	155,409	Monthly Flat Fee	0.00
NOVEMBER 2020	44,861,752	0.65	155,409	Monthly Flat Fee	0.00
DECEMBER 2020	45,954,140	0.65	155,409	Monthly Flat Fee	0.00
JANUARY 2021	44,480,277	0.65	155,409	Monthly Flat Fee	0.00
FEBRUARY 2021	32,536,262	0.65	155,409	Monthly Flat Fee	0.00
MARCH 2021	45,964,136	0.65	155,409	Monthly Flat Fee	0.00
Billing Total for Period	238,396,196		932,455		0.00

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TOTAL UPLIFT	Quantity	Unit Price	Amount
Uplift by Indirect Accounts Associated with Evidence	24,770	200.00	4,954,000
Uplift by Identifying Confirmed Fraud Accounts	1,696	200.00	339,200
Uplift by Preventing Real-time Fraudulent Transactions	51,512	5.70	293,766
Uplift by Reduced Review Costs	53,954	4.85	261,677
Uplift by Reduced Technology Costs	56,794	1.00	56,794
Uplift Over 6 Months			\$ 5,905,437

Average Cost of Fraud per Transaction

Calculated based on the average cost of fraud per account provided, the number of accounts found during the analysis window that had evidence placed on them directly, and divided by the number of transactions performed by those accounts. This method spreads the fraud losses evenly across each transaction where a loss occurred.

Accounts with Direct Evidence in Period

X Average Cost of Fraud per Account
 ÷ # of Transactions they Performed
 = Average Cost of Fraud per Transaction

ROI: Current Utilization

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**Total Savings
with TruValidate:**

\$963,081

[Total Uplift] – [Total Costs]

The Formula for Calculating ROI

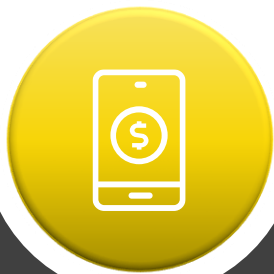
Credit One Monetary Uplift



Total Uplift

\$5,905,437

Quantifiable value of the service based on your actual traffic, unique fraud catch it provided, and reduced fraud costs



Total Investment

\$4,942,356

Total monetary value of transaction costs, one-time and monthly fees, and internal costs associated with using the service



Bottom Line ROI

\$963,081

[Uplift] – [Investment]

—

=

Overall Uplift of TruValidate Device Solution

Based on Activity, Fraud Costs, and Evidence Placement



26K+

Confirmed Fraudulent
Accounts Identified



1.2:1

6-Month ROI Ratio



\$963K

Bottom Line Savings with
TruValidate



99.6%

Accuracy Rate in
Identifying Risk



0.2%

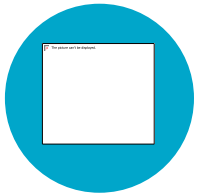
False Positive
Rate

Recommendations



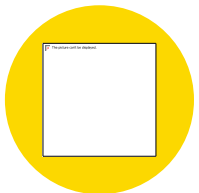
Tune business rules to target more recent fraud behaviors and reduce false negative rate

Last rule change was 4 months before analysis. Only 3 changes made over the 6-month analysis window.



Stop customer's journey for returning fraudsters with a real-time DENY

Application Fraud (1st Party) and ATO were the top 2 types of fraud identified by others first but not producing a real-time DENY.



Reduce technology costs by preventing calls to additional fraud prevention services for bad actors